

Product Features Blueprint

PhantomBuster Gap Analysis · Feature Spec · Open Source Stack · Build Priority Matrix

PRD INPUT

PRODUCT STRATEGY

FEATURE SPEC

OSS RESOURCES

Document	Product Features Blueprint v1.0
Author	Nihal Mishra — Markeye
Date	May 17, 2026
Purpose	Define features that beat PhantomBuster; drive PRD creation
Next Action	Convert each Feature Card into GitHub Issues / Linear tickets

01 — Executive Summary: Why We Can Beat PhantomBuster

PhantomBuster is a \$69–439/mo SaaS used by 2.5M people. It works — but it has deep, systemic gaps that real users complain about repeatedly across G2, Capterra, Reddit, and Trustpilot. These aren't edge cases. They are structural limitations baked into PB's architecture that they CAN'T fix without rebuilding. We can build around ALL of them.

Gap Category	Severity	% of User Complaints	Our Advantage
Account ban / datacenter IPs	CRITICAL	~40%	Residential proxies built-in
No custom workflow builder	HIGH	~25%	Visual n8n-style flow editor
Unpredictable pricing / hour burnout	HIGH	~20%	Unlimited executions, flat fee
Poor support & vague errors	HIGH	~15%	Real-time run logs + alerts
No AI personalization built-in	MED	~10%	Claude-native message gen
No multi-account management	HIGH	~18%	Account rotation built-in
Broken scripts after platform updates	HIGH	~12%	AI-powered selector recovery
No CRM / reply tracking	MED	~10%	Supabase CRM + reply inbox
Agency white-label impossible	MED	~8%	Multi-tenant from day 1

■ Opportunity: PhantomBuster's Trustpilot score is only 3.8/5 vs G2's 4.4/5. The gap = non-technical users hate it. Our product must be SIMPLER and SAFER by design.

02 — Feature Gap Cards (Each = One PRD Epic)

Each card below is a PhantomBuster gap → your product's solution. Priority = revenue/adoption impact. Effort = engineering days.

F-01: Residential Proxy Engine

Priority: **HIGH** Effort: **MED**

■ **PB Gap:** Runs on datacenter IPs. LinkedIn detects and bans. Users lose accounts. PB puts ban risk 100% on user.

✓ **Our Build:** Built-in residential proxy routing via Decodo/Smartproxy API. Auto-rotate per session. User never touches proxy config.

F-02: Unlimited Execution Model

Priority: **HIGH** Effort: **LOW**

■ **PB Gap:** Charged by execution hours (20–300 hrs/mo). Burns out mid-campaign. No rollover. Forces upgrades constantly.

✓ **Our Build:** Flat monthly fee OR per-campaign pricing. No hourly caps. Run 24/7. User pays for infra, not clock.

F-03: Visual Workflow Builder

Priority: **HIGH** Effort: **HIGH**

■ **PB Gap:** Pre-built Phantoms only. Can't build custom chains without technical knowledge. Rigid slot-based system.

✓ **Our Build:** Drag-and-drop workflow builder (n8n-style). Connect any scraper → enricher → sender → CRM in visual UI.

F-04: AI-Native Message Personalization

Priority: **HIGH** Effort: **MED**

■ **PB Gap:** AI credits as add-on. Generic template system. Blank fields when data missing. No ICP scoring.

✓ **Our Build:** Claude API baked into every outreach step. Profile scraped → AI scores ICP fit → writes custom message → sends.

F-05: Multi-Account Rotation

Priority: **HIGH** Effort: **MED**

■ **PB Gap:** One account per cookie set. No account rotation. Agencies hit limits immediately. No safe volume scaling.

✓ **Our Build:** Add N LinkedIn accounts. Campaign auto-rotates senders. 5 accounts × 20 req/day = 100/day safely.

F-06: Real-Time Run Logs & Error Alerts

Priority: **HIGH** Effort: **MED**

■ **PB Gap:** Vague error messages. Users don't know why a Phantom failed. Support slow (18+ hr response times).

✓ **Our Build:** Live terminal-style logs per run. Slack/email alert on failure. Auto-retry with exponential backoff. Clear error codes.

F-07: Smart Session Management

Priority: **HIGH** Effort: **LOW**

■ **PB Gap:** li_at cookie expires daily. User must manually refresh. Session loss = campaign crash. No auto-recovery.

✓ **Our Build:** Auto-detect session expiry (401 response). Alert user. Queue resumes when re-authenticated. Session stored in Redis.

F-08: Built-in CRM + Reply Inbox

Priority: **MED** Effort: **HIGH**

■ **PB Gap:** No reply tracking. No lead status management. No inbox. Must export CSV and manage elsewhere.

✓ **Our Build:** Supabase-powered lead CRM. Status pipeline (sent → connected → replied → qualified → booked). Unified reply inbox.

F-09: Account Warm-Up Mode

Priority: **HIGH** Effort: **LOW**

■ **PB Gap:** Zero warm-up system. New accounts hit full automation immediately. LinkedIn flags instantly.

✓ **Our Build:** Day 1-14 auto-ramp: 5→10→20→50→80 actions/day. Mimics organic growth pattern. Warm-up score displayed.

F-10: AI-Powered Selector Recovery

Priority: **MED** Effort: **HIGH**

■ **PB Gap:** Phantoms break when LinkedIn changes DOM. User stuck until PB releases patch (days/weeks).

✓ **Our Build:** AI reads updated page structure → auto-updates CSS selectors → campaign continues. Zero manual intervention.

F-11: White-Label Multi-Tenant

Priority: **HIGH** Effort: **HIGH**

■ **PB Gap:** No white-label. One workspace per account. Agencies can't manage client accounts separately.

✓ **Our Build:** Multi-tenant from day 1. Each client = isolated workspace. Agency master dashboard. Custom branding per tenant.

F-12: Behavior Trigger Campaigns

Priority: **MED** Effort: **HIGH**

■ **PB Gap:** Static campaign triggers only. Can't react to lead behavior (post liked, job change, company news).

✓ **Our Build:** Trigger campaigns on: new job post, LinkedIn post by target, comment on ICP content, company funding news.

F-13: Email + LinkedIn Waterfall

Priority: **MED** Effort: **MED**

■ **PB Gap:** LinkedIn only per campaign. Email enrichment separate. No unified sequence across channels.

✓ **Our Build:** Single campaign: LinkedIn connect → 3d wait → email follow-up → 7d wait → LinkedIn DM. One setup.

F-14: Safety Score Dashboard

Priority: **MED** Effort: **MED**

■ **PB Gap:** No visibility into account health. Users have no idea if they're close to a ban until it happens.

✓ **Our Build:** Live safety score per account (0-100). Tracks: daily action volume, acceptance rate, reply rate, IP consistency.

F-15: One-Command Docker Deploy

Priority: **HIGH** Effort: **LOW**

■ **PB Gap:** Cloud-only SaaS. No self-hosted option. Data lives on PB servers. Agencies can't white-label infra.

✓ **Our Build:** docker compose up → full stack running in 3 minutes. Client owns their data. VPS deploy guide included.

03 — Feature Priority Matrix (Build Order)

Ship order based on: revenue impact × effort. Phase 1 = MVP. Phase 2 = moat. Phase 3 = product.

Feature	ID	Phase	Effort (days)	Revenue Impact	Blocker?
Residential proxy engine	F-01	Phase 1	1–2	■ Critical	YES
Unlimited execution model	F-02	Phase 1	0.5	■ Critical	YES
Smart session management	F-07	Phase 1	1	High	YES
Account warm-up mode	F-09	Phase 1	1	High	YES
One-command Docker deploy	F-15	Phase 1	1	High	YES
AI message personalization	F-04	Phase 2	2–3	■ Critical	NO
Multi-account rotation	F-05	Phase 2	2	High	NO
Real-time logs & alerts	F-06	Phase 2	1–2	High	NO
Safety score dashboard	F-14	Phase 2	2	Med	NO
Built-in CRM + reply inbox	F-08	Phase 3	3–4	High	NO
Visual workflow builder	F-03	Phase 3	5–8	■ Critical	NO
White-label multi-tenant	F-11	Phase 3	4–6	High	NO
Email + LinkedIn waterfall	F-13	Phase 3	3–4	Med	NO
Behavior trigger campaigns	F-12	Phase 4	5+	Med	NO
AI selector recovery	F-10	Phase 4	4–6	Med	NO

■ Phase 1 (5–6 days total) ships a working, safe, self-hosted tool. That alone beats PhantomBuster on the 3 things users hate most: bans, pricing, inflexibility.

04 — Free & Open Source Resources

Everything below is free to use, fork, or reference. Organized by layer of the stack.

4.1 — Core Browser Automation (The Engine)

Repo / Tool	Language	Stars	What to use it for	Link
OpenOutreach	Python	★ Active	PRIMARY FORK — LinkedIn automation + DjangoCRM + Playwright + AI discovery	github.com/eracle/OpenOutreach
Playwright	Python/JS/TS	★ 70k+	Browser automation engine. Cross-browser. Better than Puppeteer for Python.	github.com/microsoft/playwright
puppeteer-extra-stealth	Node.js	★ 7k+	Stealth plugin collection. Defeat navigator.webdriver, canvas, CDP detection.	github.com/berstend/puppeteer-extra
rebrowser-patches	Node.js	★ 1k+	2025 stealth patches for Chromium — more current than puppeteer-extra-stealth.	github.com/rebrowser/rebrowser-patches
browser-use	Python	★ 40k+	AI agent controls real browser. Give it a task in English → it figures out the steps.	github.com/browser-use/browser-use
Stagehand	TypeScript	★ 10k+	LLM-native browser framework. Claude/GPT reads page → decides actions. Node.js.	github.com/browserbase/stagehand
Crawlee (Python)	Python	★ 5k+	Reliable crawler + Playwright integration. Built by Apify. Proxy rotation built-in.	github.com/apify/crawlee-python
inb (LinkedIn Voyager API)	Python	★ 600+	Uses LinkedIn's internal Voyager API — more stable than DOM scraping.	github.com/joshiayush/inb

4.2 — Lead Generation & Scraping

Repo / Tool	Platform	Free?	Notes
Apify Crawlee (Python)	Multi	Yes (OSS)	Full scraping framework. Playwright+HTTP. Proxy rotation. Used by thousands in prod.
Apify Store Actors	Multi	Free tier (\$5 credits)	1000+ pre-built scrapers. LinkedIn, Maps, Instagram, Apollo. Clone the concept.
SalesGPT	Email/SMS/Voice	Yes (OSS)	LLM-powered AI SDR agent. LangChain. Good for reply handling layer.
apollo-scraper (Apify)	Apollo.io	Free Actor	Scrape Apollo.io contact lists — Apollo-alternative without paying Apollo.
linkedin-scraper (Python)	LinkedIn	Yes (OSS)	Basic LinkedIn profile scraper. Voyager API. Good for email extraction.

Repo / Tool	Platform	Free?	Notes
google-maps-scraper	Google Maps	Yes (OSS)	50+ data points from GMaps. Phone, email, reviews. Local lead gen.
Crawl4AI	Any website	Yes (OSS)	AI-powered web crawler. Extracts structured data. LLM-ready output.
Hunter.io (free tier)	Email	50 req/mo free	Email finder from domain/name. API-first. Waterfall step.
Apollo.io (free tier)	B2B contacts	50 exports/mo	275M contacts. Email + phone. Free tier for enrichment.
Anymailfinder	Email	Pay-per-find	Verified email discovery. ~\$0.01 per found email.

4.3 — Infrastructure & Orchestration

Tool	Role	Free?	Why use it
n8n (self-hosted)	Workflow orchestration	Yes (OSS)	Already deployed at n8n.markeye.space. Webhook triggers, campaign scheduling.
Supabase	Database + Auth + API	Free tier	Postgres + REST API + auth. Lead CRM backend. Already on your stack.
Upstash Redis	Session/cache store	Free tier	Store li_at cookies, session state, job queues. Serverless Redis.
BullMQ	Job queue	Yes (OSS)	Node.js async job queue. Retry logic, priority, delay. LinkedIn action pacing.
Celery + Redis	Job queue (Python)	Yes (OSS)	Same as BullMQ but Python. Use if building Python-first.
FastAPI	API layer	Yes (OSS)	REST API for triggering campaigns, reading results. Already know it.
Docker + Compose	Container orchestration	Yes	One-command deploy. playwright + fastapi + redis + n8n in one compose file.
Caddy	Reverse proxy + HTTPS	Yes (OSS)	Already running on your DO droplet. Auto HTTPS for dashboard.
Hetzner CX21	VPS	\$4.15/mo	2 vCPU, 4GB RAM. Enough for Playwright + API + n8n. Cheaper than DO.
Browserless.io	Managed Chrome	\$19/mo	Hosted headless browser. Scale Playwright sessions without managing Chrome.

05 — People Currently Building This (Learn From Them)

Real builders doing this right now. Study their approaches, fork their code, follow their posts.

Builder / Project	What They Built	Status	Where to Find
eracle / OpenOutreach	Self-hosted LinkedIn automation. Playwright + DjangoCRM + LLM discovery. Exactly what we're building.	Active (2025)	github.com/eracle/OpenOutreach
joshiayush / inb	Python LinkedIn bot using Voyager API (internal LinkedIn API, more stable than DOM).	Active	github.com/joshiayush/inb
filip-michalsky / SalesGPT	AI Sales Agent. LangChain + GPT. Voice, email, SMS, WhatsApp outreach. 2.2k stars.	Active	github.com/filip-michalsky/SalesGPT
bearconnect.io (IH story)	Bootstrapped LinkedIn automation SaaS from scratch. Reached \$1.3K MRR. Good case study.	Live product	indiehackers.com (search bearconnect)
Apify Team	Largest OSS scraping ecosystem. 1000+ actors. LinkedIn, Maps, Apollo scrapers. Study the Actor architecture.	Very Active	github.com/apify apify.com/store
Sami Halawa / HuggingFace blog	Ranked top 30 OSS lead generation + outreach repos. Excellent comparison of approaches.	2025 post	huggingface.co/blog/samihalawa/automating-lead-generation-with-ai
Dux-Soup team	300K users. LinkedIn automation. 2026 roadmap: multi-account, behavior triggers, CRM native. Watch their roadmap.	Competitor	dux-soup.com/blog
HeyReach	Multi-account LinkedIn. Agency-first. \$4.8/5 G2. Study their UX patterns — it's what users WANT.	Competitor	heyreach.io
n8n community	Pre-built LinkedIn outreach + OpenAI workflow templates. Free. Shows what the market wants automated.	Active community	n8n.io/workflows (search linkedin)

Key Lessons From These Builders

- ✓ OpenOutreach proves the core stack works. Fork it instead of starting from scratch.
- ✓ SalesGPT proves AI-native outreach is the differentiator users actually pay for.
- ✓ HeyReach's success (\$4.8 G2) = multi-account + agency dashboard is THE missing feature in PB.
- bearconnect.io hit \$1.3K MRR quickly — proving the market pays for simpler LinkedIn automation.
- Apify's Actor marketplace model = future product layer (each 'Phantom' = a deployable Actor).
- ✗ Builders who start with DOM scraping hit LinkedIn blocks fast. Voyager API or residential proxies are non-negotiable.

06 — Complete Free Tools Cheat Sheet

Every tool/resource below is free or has a meaningful free tier. Use these to build v1 with near-zero cost.

Tool	Category	Free Limit	Use Case in Our Product
OpenOutreach	Core fork	Fully free (MIT)	Base LinkedIn automation engine
Playwright	Browser engine	Fully free	Run LinkedIn sessions headlessly
puppeteer-extra-stealth	Stealth	Fully free	Avoid LinkedIn bot detection
browser-use	AI browser agent	Fully free (OSS)	Intelligent multi-step flows
Crawlee (Python)	Scraping framework	Fully free	Robust crawler with proxy support
n8n self-hosted	Orchestration	Fully free	Campaign scheduling + webhooks
Supabase	Database + Auth	Free (500MB)	Lead CRM, session storage, analytics
Upstash Redis	Cache/queue	Free (10k cmd/day)	Cookie store, rate limiter
FastAPI	API layer	Fully free	REST endpoints for all automation
Docker + Compose	Deploy	Fully free	One-command local or VPS deploy
Caddy	Reverse proxy	Fully free	HTTPS for dashboard auto
Hunter.io	Email enrichment	50 req/mo free	Email from company domain
Apollo.io	B2B contacts	50 exports/mo	Lead discovery + email
Anymailfinder	Email enrichment	5 free/mo	Fallback email finder
2Captcha	CAPTCHA solver	\$0.50/1000 CAPTCHAs	Auto-solve if triggered (~\$0/mo low vol)
Hetzner CX21	VPS	€4.15/mo (~\$4.50)	Run entire stack (not free but cheapest)
Apify free tier	Cloud scraping	\$5/mo credits free	Run Apify actors for multi-platform
GitHub Actions	CI/CD	2000 min/mo free	Auto-test + deploy on push
Vercel	Dashboard hosting	Fully free (hobby)	Host React campaign dashboard
Groq API	LLM (fast)	Free tier (generous)	Fast message personalization
Anthropic Claude (Haiku)	LLM (quality)	Pay-per-use (~\$0.001/msg)	High-quality personalization + scoring

■ **TOTAL COST TO BUILD v1 (exc. proxy): ~\$4.50/mo VPS + \$0 everything else. Add Decodo residential proxy at \$14/mo → total \$18.50/mo operational cost.**

07 — Immediate Action Items

In order. Do these before writing a single line of product code.

TODAY

- ◆ Clone github.com/eracle/OpenOutreach → inspect codebase architecture
- ◆ Sign up Decodo proxy free trial → get API credentials
- ◆ Create Supabase project → design leads table schema
- ◆ Create Upstash Redis account (free) → get connection string

DAY 1

- ◆ Wire Decodo proxy into OpenOutreach Playwright config
- ◆ Test: visit 10 LinkedIn profiles → confirm no WebDriver flag detected
- ◆ Replace DjangoDB → Supabase. Map lead fields: name/headline/url/company/status
- ◆ Build n8n webhook: POST /trigger-campaign → starts Playwright job

DAY 2

- ◆ Add Claude Haiku API to message personalization step
- ◆ Implement daily limit enforcer (configurable, default 50 actions/day)
- ◆ Add session expiry detection + Slack alert on 401 response
- ◆ Run first real campaign: 20 connections to your target ICP

DAY 3–5

- ◆ Build basic React dashboard: campaign list, lead table, status pipeline
- ◆ Add multi-account support (store multiple li_at cookies in Redis)
- ◆ Package into Docker Compose: playwright + fastapi + redis + n8n
- ◆ Write client README + Loom walkthrough for first client delivery

Markeye Clone = PhantomBuster gaps fixed + AI brain + \$18.50/mo vs \$69–439/mo

Each Feature Card above = 1 Linear/GitHub Issue → this doc IS the PRD input

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